

AN APPROACH TO STUDY THE COVID-19 STAY-AT-HOME EFFECT ON WATER CONSUMPTION IN LONG ISLAND, NY USING TIME SERIES ANALYSIS OF GROUNDWATER LEVELS



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Abstract

This study investigates groundwater level trends as a proxy of water consumption after the quarantine orders imposed by the COVID-19 pandemic. Statistical analyses were conducted on 56 active USGS groundwater monitoring stations, and 238,000 groundwater level records were processed from 2000 to 2021 (Fig. 1). Groundwater time series data were filtered using a low-pass filter (KZ- Kolmogorov Zurbenko) to create long-term trends to recognize groundwater data behavior prior to and during the pandemic. It was hypothesized and supported by the data that water consumption increased, which yielded a decrease in groundwater levels from March 2020- March 2021.

Introduction

While water has been studied for quality, the quantity of water being used during the COVID-19 pandemic, especially given that the main prevention of spreading the disease is frequent handwashing and cleaning, has been overlooked. The overall goal is to observe the trends posed by the Covid- 19 Pandemic.

- Kalbusch, 2021 (Joinville, Southern Brazil): While commercial, industrial, and public water consumption dropped dramatically, the residential rates rose notably. The region's overall water consumption dropped because the drop in the other categories overwhelmed the increase in residential consumption.
- Sayeed et al., 2021 (Bangladesh): Data collected by web-based surveys and questionnaires indicated that residents frequently left the tap running while scrubbing hands
 - Of the 1,980 participants, 1,134 participants (57.27%) left the water running to scrub their hands: each wasted approximately 12.79 times more water per day than they had before frequent hand washing protocols.

Methodology

- Download and USGS data cleaning, figures, map, using R programming language in Rstudio.
- Spectral analysis (periodogram function by TSA library) to determine the window size of a moving average
- Moving average (30-, 90-, 365-day by kz function from kza library)
- Trend analysis from the overall averaged data to clarify trend(s) that occur within those frames and eliminate irregular frequencies over the entire time series dataset.

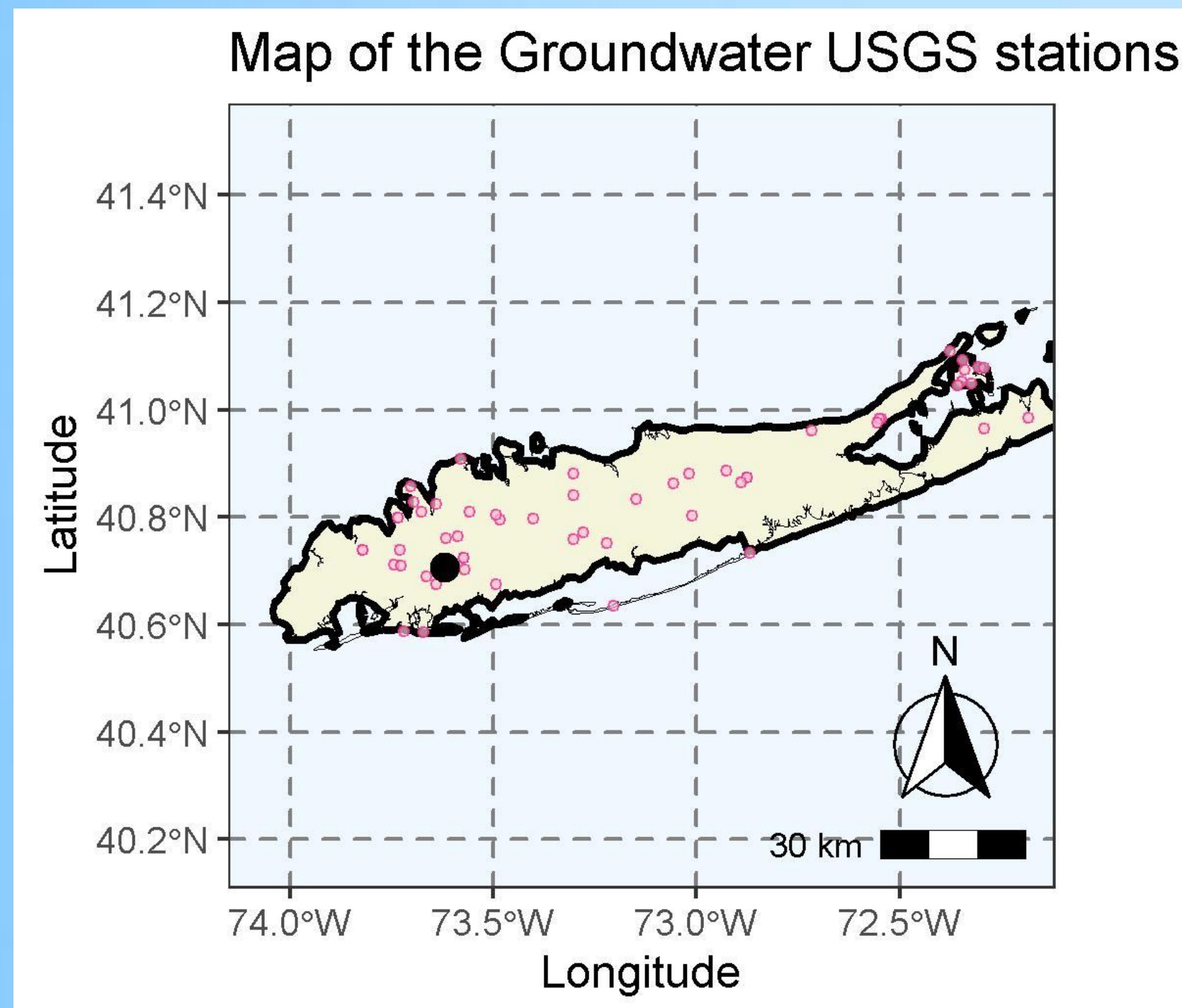


Figure 1: Map of the USGS stations monitoring groundwater levels (shown by pink dots) processed, including the stations surrounding Hempstead on Long Island (shown by the black dot).

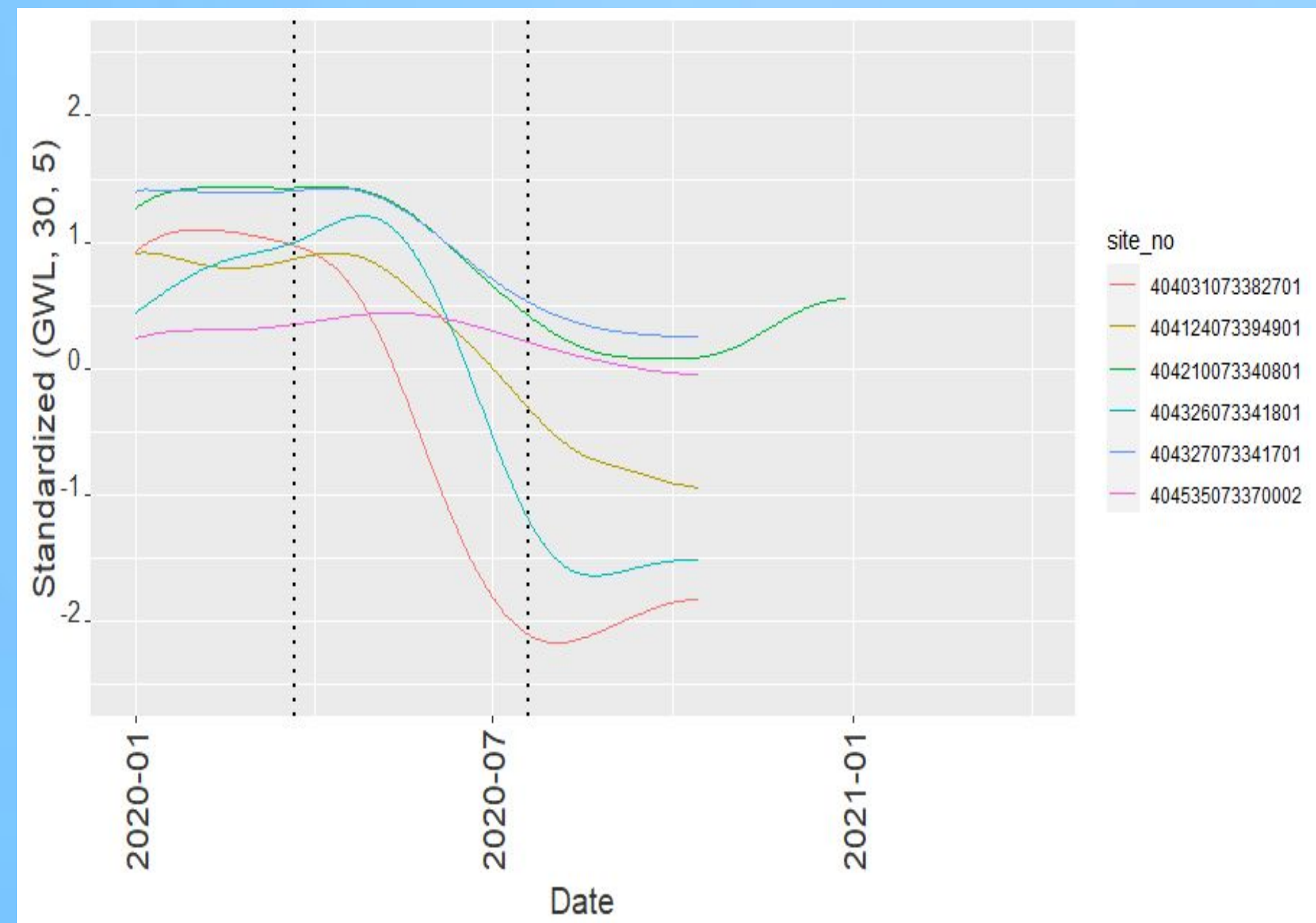


Figure 2: A prominent groundwater level decline during the interval of the stay-at-home orders imposed by the COVID-19 pandemic. Data were standardized using the “normalize()” function (method: “standardize”) after the application of kz function with a 30-day moving average and an iteration of 5 times on USGS sites near Hempstead, NY from January 2020 to March 2021.

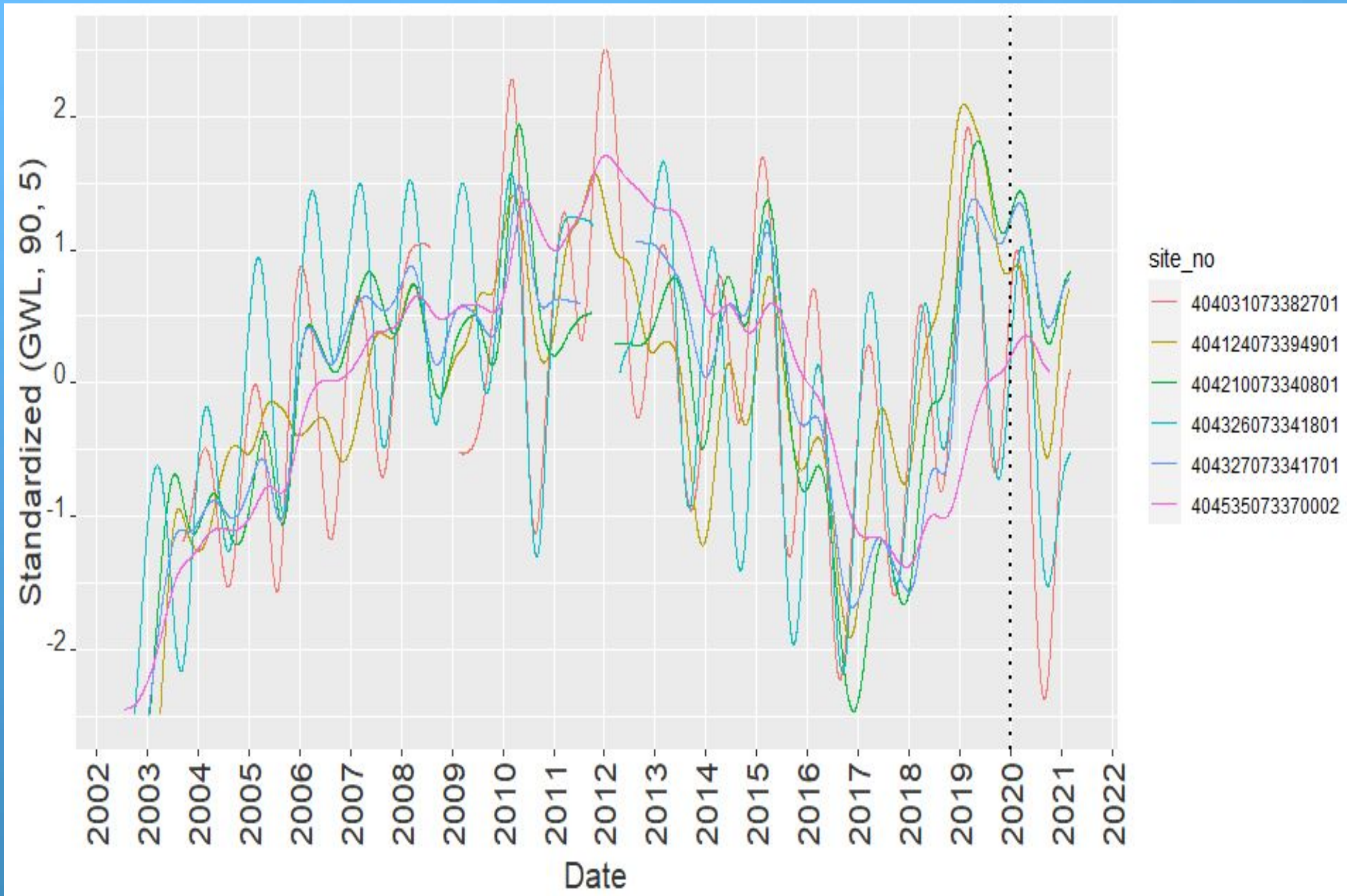


Figure 4: 90-day moving average of groundwater monitoring sites of USGS wells near Hempstead, NY. Data was processed using the "normalize()" function with a “standardize” method: displays standardized values of GWL levels after application of the kz function for a window of 90 days, iterated 5 times.

Kolmogorov-Zurbenko (KZ) low-pass filter

The KZ filter is a low-pass filter, and it has been applied with three repeated iterations (p) of a simple moving average of (m) points. The moving average of the KZ filter is provided by expression (1):

$$Y_t = \frac{1}{m} \sum_{j=-k}^k (X_{t+j}) \quad (1)$$

where $m = 2k + 1$. The output of the first iteration becomes the input for the second iteration, and so on. The time series is produced by p iterations of the filter described above, and it is denoted by the following expression (2):

$$Y_t = KZ_{m,p}(X_t) \quad (2)$$

The KZ filter has been applied, which allows a physical interpretation of the revealed trends. In addition, the KZ filter provides the minimum square of errors between the components of the time series data (Zurbenko and Sowizral, 1999).

Figure 6: Brief description of Kolmogorov-Zurbenko (KZ) low-pass filter provided by Dr. Tsakiri.

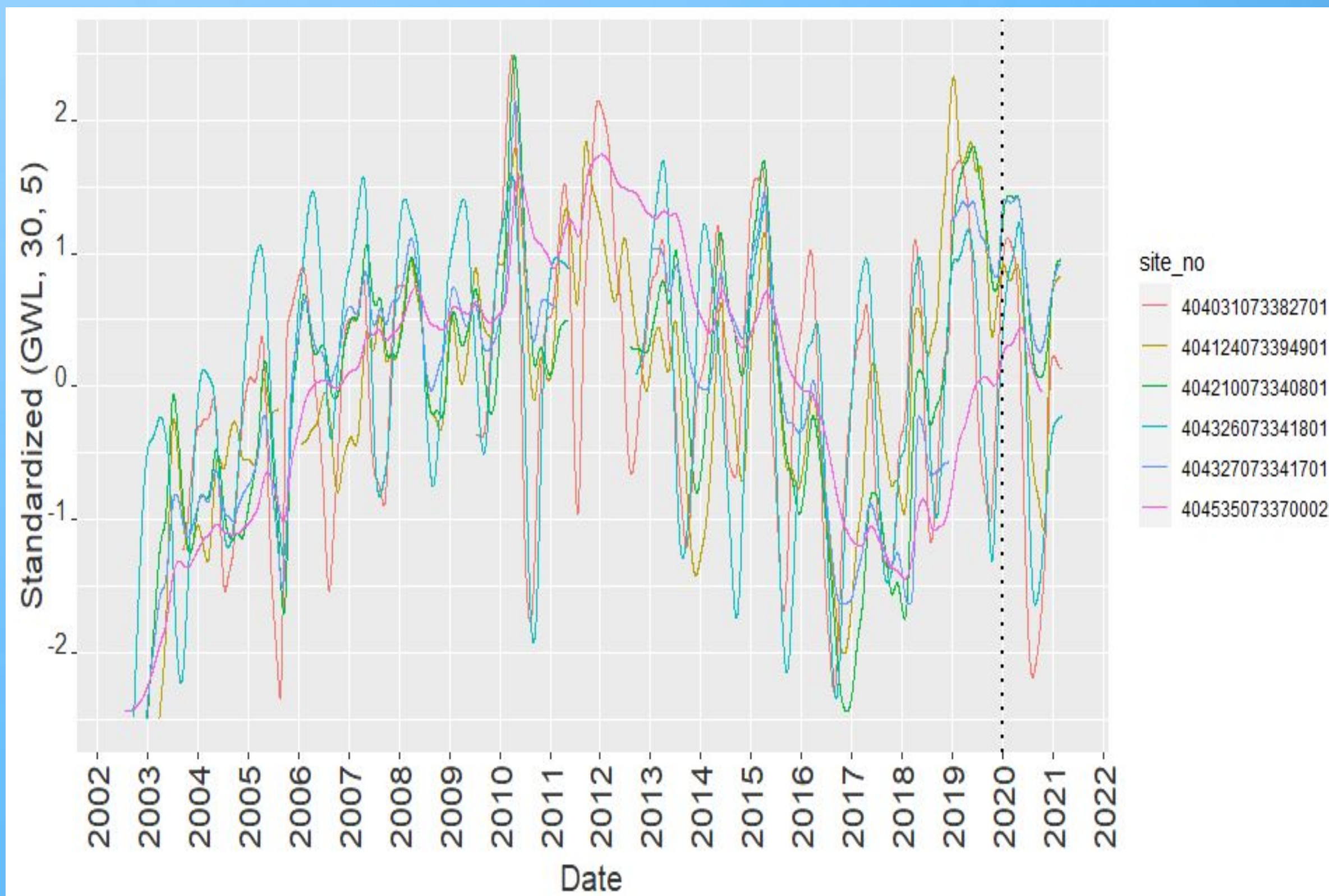


Figure 3: 30-day moving average of groundwater monitoring sites of USGS wells near Hempstead, NY. Data was processed using the "normalize()" function with a “standardize” method: displays standardized values of GWL levels after application of the kz function for a window of 30 days, iterated 5 times.

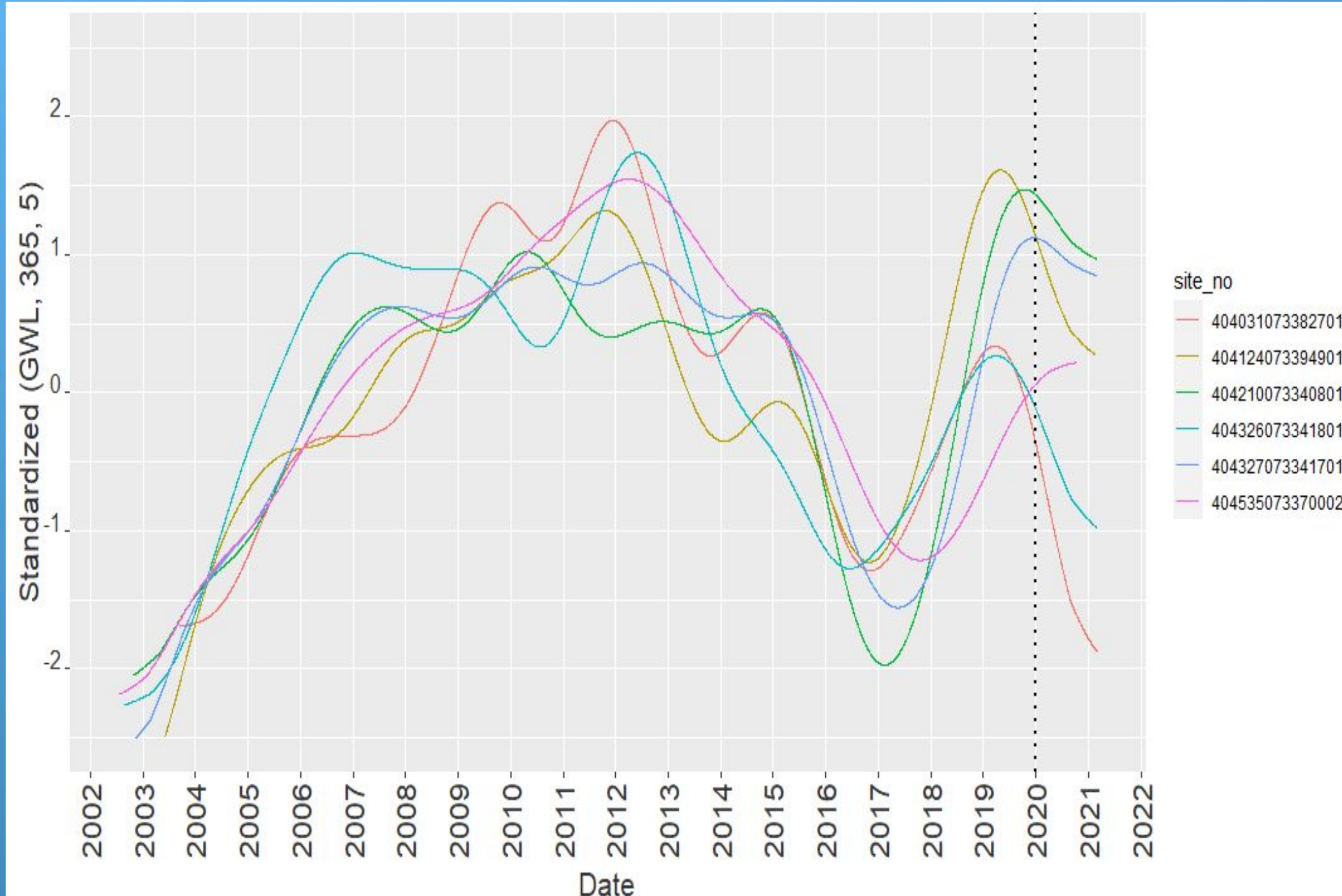


Figure 5: 365-day moving average of groundwater monitoring sites of USGS wells near Hempstead, NY. Data was processed using the "normalize()" function with a “standardize” method: displays standardized values of GWL levels after application of the kz function for a window of 365 days, iterated 5 times.

Results

- Moving average graph of a 30-day moving average from Jan. 2020 to March 2021 shows a prominent groundwater level decline during the stay-at-home orders imposed by COVID-19 regulations (Fig. 2).
 - 2003: low, start of most of the stations' data.
 - 2012: peak in groundwater trends.
 - 2017: secondary low.
 - 2018: secondary implied rise disturbed by fall in early 2020 (Fig. 3).
- Kz filter for the 90- and 365- day moving average mirrors the previous 30-day average, with peaks in 2012, and a drop in 2017 and a subsequent dip in early 2020 (Fig. 4, 5 & 6).

Discussion

- Groundwater supply management: during emergencies, overpumping can occur, so groundwater must be protected and used sustainably.
- Short-term and long-term assessments: confirm a significant drop in groundwater levels in correlation with the events of COVID-19.
- Future analysis: the observed trends found in these graphs may pose as a statistical outlier for future research into groundwater data in and around Long Island (Fig 4 & 5).

Conclusion

The groundwater levels in the 30-, 90-, and 365-moving averages may imply water consumption in Hempstead increasing during the COVID-19 lockdown (Fig. 2, 3, & 5).. COVID-19 shows a prominent impact on the long and short-term timescale of groundwater levels, thus far. The hypothesis that water usage would increase drastically throughout the lockdown period from the COVID-19 pandemic is supported through the downward trend and suggests that groundwater was more heavily consumed over the year 2020.

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Credit authorship contribution statement

Buchbinder, B.: data curation, R coding, writing - Introduction, Methodology, Conclusion, editing; Phan, E.: R coding, data formatting & formal analysis, writing - Introduction, Methodology, Conclusion, Discussion, editing; Renner, F.: writing - Abstract, Introduction, Conclusion, Discussion, editing; Liebowitz, A.: reference citing, editing; Achek, M.: writing - Conclusion, Discussion, editing; Marsellos, A.E.: Supervision, Conceptualization, R coding, editing; Tsakiri, K.G.: R coding, writing - Methodology